THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** conservation of mass	Mass is neither created nor destroyed
2	**(a)** Student A is correct	Na_2CO_3 is soda ash
3	**(d)** 2	$CaSO_4 \cdot 2\,H_2O$
4	**(c)** aluminium	$Al_2O_3 \cdot 2\,H_2O$
5	**(d)** hydrogen	$Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$
6	**(b)** Incorrect — it does not ionise	Covalent compound
7	**(b)** catenation and tetravalency	Two key properties
8	**(b)** diffusion	Body surface
9	**(a)** central nervous system	CNS
10	**(c)** kala-azar	Protozoan
11	**(c)** Spirogyra	Filamentous alga
12	**(a)** 1/2	50%
13	**(c)** shorter plants	Less hormone
14	**(a)** both concave mirror and concave lens	Negative f
15	**(d)** atmospheric refraction	Bending
16	**(a)** volt	J/C
17	**(d)** $R/4$	Each $R/2$, in parallel
18	**(b)** 10 A	$I \approx 9$ A
19	**(b)** ozone layer	Stratosphere
20	**(c)** Incorrect — it is non-biodegradable	Persists

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	An unbalanced equation, e.g. $Mg + O_2 \longrightarrow MgO$	1+1
22	Black CuO dissolves forming blue-green copper chloride: $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$	1+1
23	They need high energy to maintain constant body temperature, so efficient oxygen supply is required.	2
24	Thyroxine; goitre	1+1
25	If the egg is not fertilised, the thick uterine lining is not needed and breaks down.	2
26	Gregor Johann Mendel; garden pea	1+1
27	$v = c/n = 2 \times 10^8$ m/s	2
28	The magnetic field of the conductor interacts with the external magnetic field.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	M = iron; oxide Fe_3O_4; $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$	1+1+1
30	To protect it from rusting; even if scratched, zinc being more reactive corrodes first, protecting iron.	1+2
31	Bony skull (cranium) and a fluid-filled balloon (cerebrospinal fluid) act as shock absorbers; the brain is delicate and vital	2+1
32	An outgrowth (bud) develops due to repeated cell division, grows into a tiny individual and detaches to live independently	3
33	Air molecules are smaller than the wavelength of visible light, so they scatter shorter-wavelength blue light more strongly; the scattered blue light enters our eyes	3
34	Current produces a magnetic field which acts on the needle; when switched off, needle returns to N–S	3

35	Stomach HCl aids digestion; excess acid causes pain and irritation; antacids (mild bases like milk of magnesia) neutralise it.	1+1+1
	OR: (i) sodium carbonate decahydrate $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$ — cleaning (ii) sodium hydrogencarbonate $\mathrm{NaHCO_3}$ — antacid (iii) calcium oxychloride $\mathrm{CaOCl_2}$ — disinfecting water	1+1+1
36	Break down dead organisms and wastes into simple substances that return to soil; bacteria, fungi	2+1
	OR: (i) grass → deer → lion (ii) grass → grasshopper → frog → snake	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $m = h'/h = -v/u$ (ii) $\frac{1}{v} = -\frac{1}{12} + \frac{1}{6} = \frac{1}{12} \Rightarrow v = +12$ cm (behind mirror); $m = 2$; virtual, erect, magnified	1+4
	OR: (i) $P = 1/f$ (m); dioptre (ii) $f = -0.5$ m = −50 cm, concave (iii) $P = +2$ D; $f = 50$ cm	1+2+2
38	(i) series with same functional group, successive members differ by $\mathrm{CH_2}$ (ii) methanol $\mathrm{CH_3OH}$, ethanol $\mathrm{C_2H_5OH}$, propanol $\mathrm{C_3H_7OH}$ (iii) physical properties (m.p., b.p.) change gradually with mass; chemical properties similar	1+2+2
	OR: (i) chloro/bromo prefix; –ol, –al, –one suffixes replacing final 'e' of alkane (ii) bromoethane; propanal	3+2
39	(i) $V \propto I$ at constant temperature; not valid for non-ohmic conductors/changing temperature (ii) straight line through origin; $R \approx 16.6\ \Omega$ (slope)	2+3
	OR: (i) $P = VI$, kWh $= 3.6 \times 10^6$ J (ii) daily $1.2 + 0.6 = 1.8$ kWh; monthly 54 units; ₹270	2+3
40	(i) carbohydrates: salivary amylase (mouth), pancreatic amylase; proteins: pepsin (stomach), trypsin (intestine); fats: bile emulsifies, lipase digests (ii) villi of the small intestine	4+1
	OR: (i) nostrils → pharynx → larynx → trachea → bronchi → bronchioles → alveoli (ii) thin walls, large surface area, rich blood supply (iii) more $\mathrm{O_2}$ needed for energy	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER